TraceIQ

Product Requirements Document

Supply-chain traceability and AI-assisted recall command center

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Stack	React 19 / TypeScript / Vite | FastAPI / SQLAlchemy / Alembic

Document control

Item	Detail
Product	TraceIQ - supply-chain traceability and AI-assisted recall command center
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Status	Draft v1.0 for review
Date	30 October 2026
Basis	Current prototype (React/TypeScript frontend, FastAPI backend) and its implementation plan. Requirements marked Built already exist; Partial and Planned are open work.
Priority key	P0 = must have for first production release; P1 = should have; P2 = later

Contents

1. Overview and problem
2. Goals, non-goals and success metrics
3. Users and personas
4. User stories
5. Functional requirements
6. AI requirements
7. Non-functional requirements
8. Key user flows
9. UX and screen requirements
10. Data and integration requirements
11. Release plan
12. Risks, assumptions and open questions

1. Overview and problem

Product summary. TraceIQ gives manufacturers and distributors one place to trace any material or product lot from supplier to customer, detect record gaps automatically, and run a controlled recall. Deterministic rules decide what is true; AI only explains verified evidence; a named human must approve any recall.

The problem

- When a lot is found faulty, teams must find every finished lot, stock location, shipment and customer it touched, often within hours.
- Supporting records (purchase orders, certificates, production records, delivery proofs) live in separate documents and systems, and links between them are frequently missing.
- Quantity mismatches (more shipped than produced) and unlinked recipients are discovered during a recall, when time matters most.
- Recall progress is tracked in spreadsheets and email, so containment, customer notification and reconciliation are hard to prove afterwards.

Opportunity

A single system of record for genealogy, evidence and recall actions can shorten scoping time, make gaps visible before a recall, and produce an auditable trail for regulators and customers.

2. Goals, non-goals and success metrics

Goals

- G1. Let a user scope the full upstream and downstream impact of a lot in one query, with missing links stated explicitly.
- G2. Surface record gaps and quantity discrepancies continuously, not only during a recall.
- G3. Make recall activation, containment and closure controlled, human-approved and fully audited.
- G4. Turn supporting documents into reviewed, linked evidence with minimal manual typing.
- G5. Provide AI explanations that are grounded, clearly labelled and never authoritative.

Non-goals (this release)

- Replacing the customer's ERP, WMS or quality management system; TraceIQ complements them.
- Sending real customer notifications; notifications are recorded as trackable actions only.
- Machine-learning pipelines, graph databases or real-time streaming.
- AI-determined recall scope or autonomous action of any kind.

Success metrics (proposed targets - to be confirmed with stakeholders)

Metric	Definition	Proposed target
Time to scope	Minutes from selecting a lot to a complete impact view (lots, stock, shipments, customers)	Under 5 minutes on the full seed dataset
Planted-gap detection	Share of the four planted scenarios detected by deterministic rules	100% (missing input, over-shipment, missed customer, saleable recalled stock)
Evidence coverage	Share of recall-scope lots with at least one verified linked document	Tracked from release; target set after pilot
Closure integrity	Recalls closed with zero unresolved blockers	100% (enforced by system)
Audit completeness	Recall state changes with an audit event	100%
AI safety	AI responses that fail schema validation but are shown to users	0

3. Users and personas

Persona	Needs	Main screens
Quality / recall manager	Scope impact fast, approve recalls, track containment and customer actions, reconcile and close with proof.	Recall Command Center, Traceability, Quality, Reports
Supply-chain analyst	Keep supplier, lot, production, inventory and shipment records correct; fix flagged gaps.	Suppliers, Inventory and Shipments, Anomalies
Document reviewer	Upload evidence, confirm extracted fields, link documents to records, run checks.	Document Intelligence, Validation Center
Executive / auditor	See operational health and risk at a glance; inspect history and recall reports.	Dashboard, Audit Log, Reports
Administrator (planned)	Manage users, roles, facilities and AI settings.	Admin screens (not built)

4. User stories

ID	As a... I want to...	So that...
US-01	Quality manager: enter a lot ID and see upstream inputs and downstream lots, stock, shipments and customers	I know exactly what a recall would cover
US-02	Quality manager: simulate a recall before approving it	I can review scope with no side effects
US-03	Quality manager: approve activation with my name and a note	stock is quarantined and shipments held in one controlled step
US-04	Quality manager: see exactly what blocks closure	I only close a recall when it is genuinely complete
US-05	Analyst: see shipments with no valid recipient and fix them	downstream impact is complete before a recall happens
US-06	Analyst: run a quality check on a lot or supplier	release decisions follow versioned rules and leave evidence
US-07	Reviewer: upload a certificate or delivery proof and correct extracted fields	records are backed by reviewed evidence
US-08	Reviewer: link a document to a lot, order, shipment or recall	evidence is findable from the record
US-09	Investigator: ask a question about verified evidence	I get an explanation with stated uncertainties, clearly marked as AI
US-10	Executive: view a dashboard for the last day, week, month or quarter	I can judge operational and recall readiness quickly
US-11	Auditor: browse an append-only history of who did what	I can reconstruct any decision
US-12	Administrator (planned): assign roles and limit access by facility	only authorized people can approve recalls

5. Functional requirements

Status: Built = in the current prototype; Partial = some of the requirement exists; Planned = not yet built.

5.1 Records and master data

ID	Requirement	Acceptance criteria	Pri	Status
FR-1.1	Maintain suppliers, products, RAW/FG lots, production orders with input lots, inventory balances, shipments and customers.	Create/list/update through UI and API; invalid input returns a clear error.	P0	Built
FR-1.2	Preserve broken links rather than reject them.	Records referencing missing lots or orders can exist and appear as findings.	P0	Built
FR-1.3	Full procurement, receipt, delivery, return, BOM and inventory-movement domain.	Each entity has CRUD, links and reconciliation inclusion.	P1	Planned

5.2 Traceability and quantities

ID	Requirement	Acceptance criteria	Pri	Status
FR-2.1	Trace from a lot, order, inventory, shipment, supplier or customer in both directions.	Result lists nodes, edges, missing links and quantity findings.	P0	Built
FR-2.2	Reconcile produced quantity against inventory and shipments.	Over-allocation is flagged; units are respected.	P0	Built
FR-2.3	Account for returns, destruction and documented process losses.	Reconciliation categories cover all in the process.	P1	Planned

5.3 Quality and validation

ID	Requirement	Acceptance criteria	Pri	Status
FR-3.1	Versioned quality checks returning PASS, FAIL, PENDING or INCOMPLETE.	Each run stored with findings and evidence; none overwritten.	P0	Built
FR-3.2	FAIL or INCOMPLETE opens a corrective action and quarantines the lot.	CAPA exists and is OPEN; status change is audited.	P0	Built
FR-3.3	Persisted validation runs and findings for documents and lot/supplier checks.	Run shows pass/fail/incomplete/warning counts; findings can be resolved with actor and note.	P0	Built
FR-3.4	All workflow transitions guarded by configurable rules.	Blocking rules prevent a transition until passed.	P1	Partial

5.4 Anomaly detection

ID	Requirement	Acceptance criteria	Pri	Status
FR-4.1	Detect missing input lot, missing production order, quantity above production, shipment without recipient, recalled stock still available.	Each appears with severity, entity and explanation.	P0	Built

5.5 Recall management

ID	Requirement	Acceptance criteria	Pri	Status
FR-5.1	Simulate a recall with no side effects.	Failed simulation creates no partial case.	P0	Built
FR-5.2	Activation requires a named approver and note.	Request without either is rejected; approval stored on the case.	P0	Built
FR-5.3	Activation quarantines available stock, holds in-transit shipments and creates customer-notification actions atomically.	All happen together or not at all; audit event written.	P0	Built
FR-5.4	Prevent overlapping active recalls.	A second recall touching the same finished lots is rejected.	P0	Built
FR-5.5	Track actions, record reconciliation, evaluate closure blockers.	Closure fails while any blocker exists and lists each one.	P0	Built
FR-5.6	Send real customer notifications.	Delivery status recorded.	P2	Planned

5.6 Documents

ID	Requirement	Acceptance criteria	Pri	Status
FR-6.1	Bounded upload with duplicate detection.	Limits 15 MB, 150 pages, 10 files; duplicate checksum rejected.	P0	Built
FR-6.2	Extract text and fields from PDF, DOCX, XLSX, CSV and images with provenance.	Each field traces to its source page or snippet.	P0	Built
FR-6.3	OCR for scanned files.	Works when Tesseract is available; otherwise states OCR unavailable and requires manual review.	P1	Partial
FR-6.4	Human review before checks.	Checks cannot run on an unreviewed document.	P0	Built
FR-6.5	Link documents to business records.	One link per document-record pair; shown on both sides.	P0	Built
FR-6.6	Durable processing and object storage.	Jobs survive restart; files in private object store.	P1	Planned

5.7 Dashboard, audit and reports

ID	Requirement	Acceptance criteria	Pri	Status
FR-7.1	Executive dashboard with period selector (today, 7d, 30d, 90d).	Charts change by period; current totals labelled as current-state.	P0	Built
FR-7.2	Labelled demo preview that never writes records.	Preview clearly marked; switching to live data is one control.	P1	Built
FR-7.3	Append-only audit history.	Key mutations write events with actor and details; no edit or delete.	P0	Built
FR-7.4	Recall report export as CSV and PDF.	Report includes scope, actions, reconciliation and audit links.	P0	Partial

5.8 Identity and administration

ID	Requirement	Acceptance criteria	Pri	Status
FR-8.1	Authentication and organisations.	Only signed-in users access data.	P0	Planned
FR-8.2	Role-based access and facility scoping.	Only authorized roles can approve a recall or resolve findings.	P0	Planned
FR-8.3	Admin screens for model settings and users.	Admin can change allowed models and budgets.	P1	Planned

6. AI requirements

ID	Requirement	Status
AI-1	AI runs only on the server; the API key is never exposed to the browser or source.	Built
AI-2	Input is limited to verified, authorized evidence; its values are treated as data, not instructions.	Built
AI-3	Output must match a fixed structure (summary, evidence, uncertainties, next steps); invalid output is discarded.	Built
AI-4	AI must not decide recall scope, bypass rules, change recall state or approve action.	Built
AI-5	Every AI answer is labelled; when AI is unavailable the product says so and deterministic results remain usable.	Built
AI-6	Task-based model routing, allowed-model list, usage budgets, redacted request and usage ledger, retries and fallback provider.	Planned
AI-7	Optional AI-assisted document extraction behind strict validation and human review.	Planned
AI-8	AI-drafted recall reports for human edit.	Planned

7. Non-functional requirements

Area	Requirement	Pri
Integrity	Recall activation and production gates are transactional; failures leave no partial state. Idempotency and concurrency tests added before release.	P0
Security	Authentication, RBAC, secret management, tenant isolation, TLS, encrypted storage, retention policy before any production use.	P0
Auditability	Audit and validation history immutable; approver identity tied to an authenticated user.	P0
Performance	Interactive screens usable on the full seed dataset (about 8,000 finished lots and 20,000 shipments); browser never receives the whole dataset; collections paginated and filterable.	P0
Reliability	Background work durable across restarts; migrations versioned and run before start.	P1
Usability	Responsive layout; loading, empty, validation and error states on every screen; dark and light themes.	P0
Accessibility	Keyboard operable and readable contrast; checked before release.	P1
Portability	Runs locally on SQLite; production on PostgreSQL; Docker Compose clean start.	P0
Observability	Request IDs and structured logs.	P1
Reproducibility	Fixed-seed dataset generation with identical output across runs.	P1

8. Key user flows

Flow A - Investigate and recall a lot

- Open Traceability, enter the suspect lot; review upstream sources, downstream finished lots, stock, shipments, customers and flagged missing links.
- Open Recall Command Center and run a simulation; review affected units and gaps. Optionally ask AI Investigation to explain the evidence.
- Approve: enter approver name and note; the system quarantines stock, holds shipments and creates customer actions.
- Work the action list; record reconciliation by category.

- Run the closure check; resolve each blocker; close. Open the recall report.

Flow B - Bring a document into the record

- Upload one or more files in Document Intelligence; duplicates are rejected.
- Wait for processing; open the preview and extracted fields with source provenance.
- Correct fields and record a review decision.
- Link the document to a supplier, lot, order, shipment or recall.
- Run checks in Validation Center; resolve findings with a note.

Flow C - Find and fix gaps

- Open Anomalies; filter by severity.
- Open the related lot or shipment; correct the record (for example assign a customer to a shipment).
- Re-run the quality check; confirm the finding clears and the audit log shows the change.

9. UX and screen requirements

Screen	Must show / allow
Overview	Three KPIs; four hoverable charts (operational throughput, document processing, inventory positions, recall work); activity and risk panels; period and source (demo/live) controls.
Suppliers	Supplier list, status, risk score; create supplier; qualification check.
Traceability	Interactive upstream/downstream graph; missing-link and quantity findings.
Quality	Run checks; result history; corrective actions with resolution.
Inventory and Shipments	Balances by warehouse and status; shipments; recipient correction.
Recall Command Center	Simulation, approval form (name and note required), action tracker, reconciliation form, closure blockers.
AI Investigation	Question box; answer with evidence, uncertainties, next steps and AI label; unavailable state.
Anomalies, Audit, Reports	Filterable findings; chronological audit events; recall report view and export.
Document Intelligence	Upload with progress; preview; field correction; review; record links; processing history.
Validation Center	Rule registry; runs; findings with expected vs actual, evidence and resolution.

Global: collapsible sidebar (drawer on mobile); persistent theme choice; demo preview always visibly labelled; every list has loading, empty and error states.

10. Data and integration requirements

- **Core entities:** suppliers, products, lots, production orders and inputs, inventory, customers, shipments, quality checks, corrective actions, recalls and actions, audit events, documents and extraction/review/link/processing records, validation rules, runs and findings.
- **Dataset for testing:** deterministic seed 20261022 with about 500 suppliers, 10,000 raw lots, 5,000 production orders, 8,000 finished lots and 20,000 shipment or delivery records. Ground-truth labels kept apart from operational data.
- **Planted scenarios to detect:** a missing input lot; shipped or delivered quantity above production; a downstream recall customer that is missed; recalled inventory still saleable.
- **Dataset gaps to close:** data dictionary; BOM and delivery/return entities; year-prefixed IDs and scale parameter; isolated ground truth; integrity checks; export, manifest and repeat-run checksum.

- **Integrations (this release):** Gemini via backend only. Future: identity provider, S3-compatible storage, import jobs, email delivery for notifications.
- **Document types:** 34 built-in categories (purchase orders, invoices, certificates, inspection and production records, shipping and delivery proofs, complaints, recall and CAPA records) plus configurable additions.

11. Release plan

Milestone	Scope	Exit criteria
M0 - Prototype (current)	Records, traceability, quality, anomalies, recall workflow, document pipeline, validation engine, dashboard, basic AI, Docker Compose.	Lint and build pass; 17 backend and 3 browser tests pass; migrations clean.
M1 - Foundation hardening	Request IDs, structured logs, authentication and RBAC foundation.	Recall approval restricted to an authorized role.
M2 - Evidence at scale	Durable processing, object storage, document browser tests, optional LLM extraction with validation.	Jobs survive restart; document flow covered end to end.
M3 - Rules everywhere	Configurable transition gates; broader finding and retest coverage.	All blocking transitions driven by the registry.
M4 - AI controls	Model routing, allow-list, budgets, usage ledger, retries and fallback.	Usage and cost visible; limits enforced.
M5 - Domain and data	Procurement-to-return domain; remaining screens; dataset completion.	All four planted scenarios asserted; two-run checksum matches.
M6 - Handover	Concurrency tests, CSV/PDF reports, retention, accessibility, clean-start validation.	Feature-completion matrix signed off.

12. Risks, assumptions and open questions

Risks

Risk	Impact	Mitigation
No authentication today	Anyone with access could approve a recall	Do not expose; deliver M1 first
Incomplete source records at customers	Gaps shown as uncertainty, not answers	Explicit missing-link reporting; document evidence
AI misuse or over-trust	Wrong scope or action	Advisory only, labelled, schema-validated, no write access
In-process background jobs	Lost processing on restart	Durable queue in M2
Regulatory expectations	Prototype not a regulated system	State the boundary; harden before regulated use

Assumptions

- Customers can supply lot, production, inventory and shipment data, by import or entry.
- A named human approver exists for every recall decision.
- A single organisation per deployment for the first release.

Open questions

- Which identity provider and role model should be supported first?
- Should reports ship as CSV and PDF together in the first production release?
- Is the large dataset loaded into PostgreSQL or generated on first run?
- What retention period applies to documents and audit events?
- Which regulated industries are first targets, and what do they require?
- What are the agreed numeric targets for the success metrics?